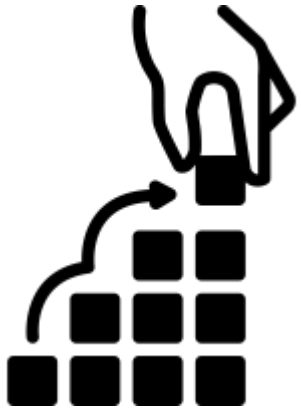


INTRODUCTION to BLOCKCHAIN

CHAPTER1 : FUNDAMENTALS OF BLOCKCHAIN

Dr. Noureddine Lasla

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WHAT IS BLOCKCHAIN?

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Abstract answer:

A Blockchain provides coordination between many parties, when there is no single trusted party

If trusted party exists → No need for blockchain

[Financial systems: often no trusted party]

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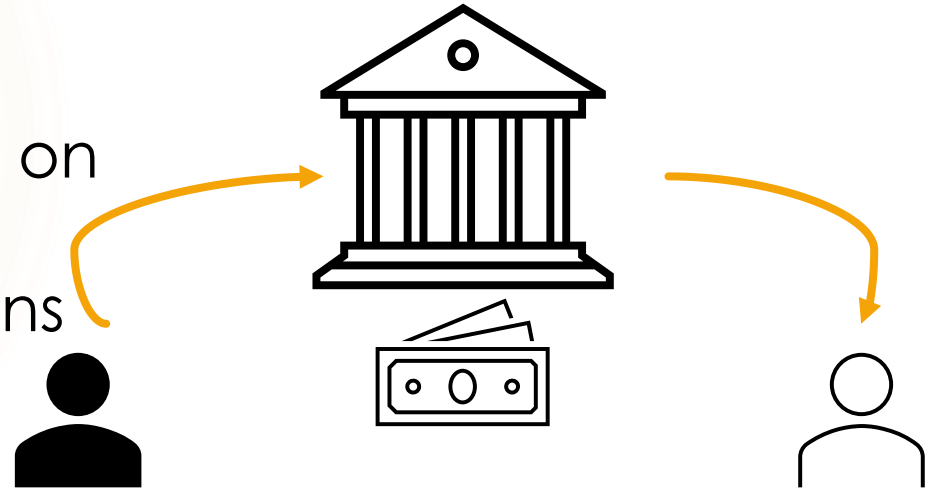
What is Blockchain? (Starting with Trust)

Why Do We Need Trust in Transactional Systems?

Many online systems, like *banking, online payments, and supply chain*, involve **multiple parties** who need to **exchange** information or transactions.

These systems depend on a **trusted third party** to

1. facilitate transactions
2. establish trust



Example: When you transfer money online, you trust the bank to **make sure** the money goes to the right person and isn't lost.

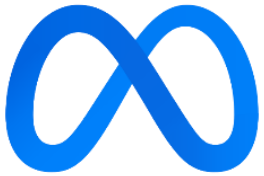
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What is Blockchain? (Starting with Trust)

Why Do We Need Trust in Transactional Systems?

Other examples of third-party facilitators:
Facebook, Gmail, …







amazon



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What is Blockchain? (Starting with Trust)

What is the Problem with Centralized Trust?

These systems rely on one central authority to validate and control everything (like a bank or a company server).

- **Issues**:
  - **Single Point of Failure**: If the central authority fails, everything fails (e.g., a bank system going offline).



  - **Power Misuse**: The central authority has control over everything and can decide what to allow or block.



  - **Vulnerable to Attacks**: If the central system is hacked, all the information can be stolen or altered.



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What is Blockchain? (Starting with Trust)

What is the Problem with Centralized Trust?

- Bank systems going down → leading to **lack of access to funds**.
- Data breaches in centralized servers → **exposing sensitive information**.

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What is Blockchain? (Starting with Trust)

What is the Problem with Centralized Trust?



- 2008 Financial Crisis and Bank Failures:
  - Centralized banks and financial institutions played a major role in the 2008 global financial crisis, caused by **risky lending practices and poor management of financial products**.
- Facebook-Cambridge Analytica Scandal (2018):
  - Cambridge Analytica harvested the personal data of **millions of Facebook users without their consent** and used this data to influence political campaigns, including the 2016 U.S. Presidential election.
- Google has been tracking Android users even with location services turned off (2017)



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CAN BLOCKCHAIN SOLVE THE TRUST PROBLEM?

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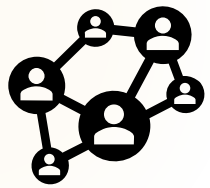
YES!!
BUT HOW?

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Blockchain as Solution to Trust problem

How Blockchain Solves the Trust Problem?

- **Decentralization**: Instead of trusting one central authority, blockchain spreads trust across many participants (a **network** of computers).



- **No Single Point of Control**: No single entity controls the system, so it's harder to fail or be attacked.



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Blockchain as Solution to Trust problem

More….

- **Everyone Can See the Data:**
  - This transparency helps people trust the system because everything is visible and verifiable.



- **Consensus:**
  - Decisions are made by agreement (consensus) from multiple participants, not just one authority.



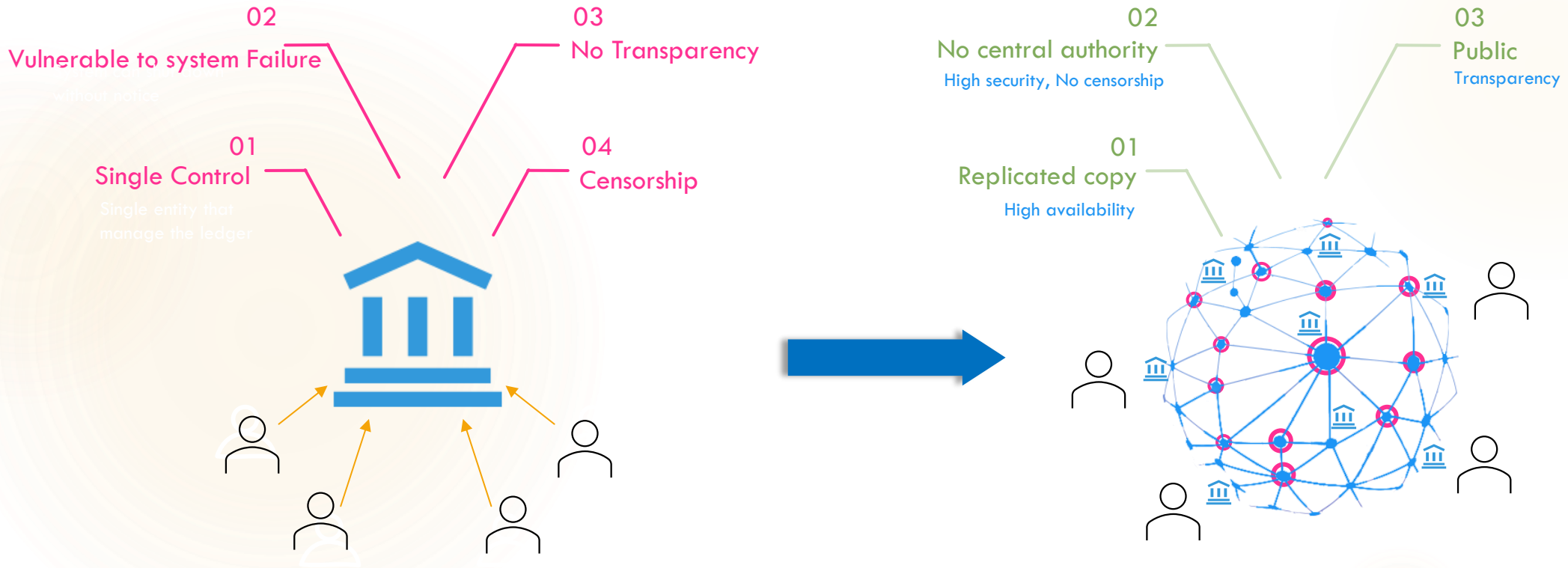
- **Tamper-Proof:**
  - Once something is recorded on the blockchain, it cannot be changed.



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Centralized vs. Decentralized Systems

Key Differences



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Centralized vs. Decentralized Systems

Key Differences

Centralized System	Decentralized System (Blockchain)
Single entity controls the system	Control is distributed across many participants
Vulnerable to system failure	More resilient as many nodes store the data
Lack of transparency	All transactions are public and verifiable
Censorship possible	No single party can block or alter transactions

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Key Characteristics of Blockchain (1/4)

1- Decentralization (No Central Authority)



- Blockchain operates on a network of nodes (computers) without a central controlling entity.
- How It Works: Every participant (node) in the network holds a copy of the entire blockchain and helps verify transactions.
- Benefit: This makes the system more resilient and prevents a single point of failure or control.

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Key Characteristics of Blockchain (2/4)

2- Transparency (Public Ledger)



- Every transaction on the blockchain is recorded in a public, immutable ledger.
- Visible to All: Anyone in the network can view and verify transactions.
- Benefit: This ensures accountability and trust, as nothing can be hidden or altered without the network knowing.

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Key Characteristics of Blockchain (3/4)

3- Immutability (Permanent Records)



- Once a transaction is added to the blockchain, it cannot be altered or deleted.
- How It Works: Transactions are grouped into blocks, which are cryptographically linked. Changing a block would require altering every subsequent block, which is nearly impossible.
- Benefit: This makes blockchain tamper-proof and reliable for storing important data

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Key Characteristics of Blockchain (4/4)



4- Security (Cryptographic Protection)

- Blockchain uses advanced cryptographic techniques (e.g., hashing, digital signatures) to secure data and ensure transaction integrity.
- Proof-of-Work: A consensus mechanism like PoW ensures that transactions are validated through computational puzzles, making it difficult for bad actors to manipulate the system.
- Benefit: This makes blockchain highly secure and resistant to fraud or unauthorized changes.

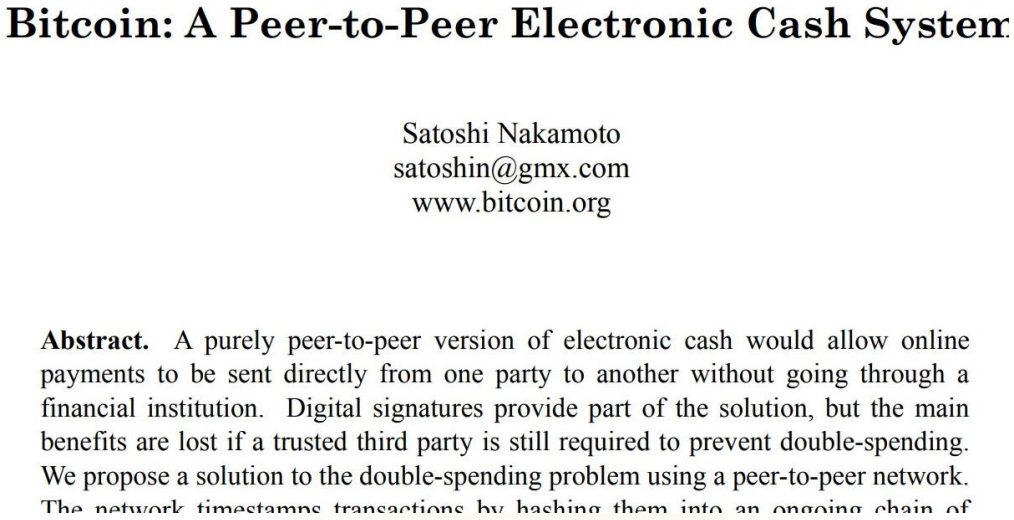
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HISTORY OF BLOCKCHAIN

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History of Blockchain

The Birth of Blockchain



Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto
satoshin@gmx.com
www.bitcoin.org

Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of

- **Bitcoin Whitepaper (2008):**
  - Blockchain technology was introduced by **Satoshi Nakamoto** in the Bitcoin whitepaper titled "*Bitcoin: A Peer-to-Peer Electronic Cash System.*"
  - **Purpose**: To create a decentralized, peer-to-peer digital currency (Bitcoin) that doesn't rely on banks or governments for transactions.
  - **Key Innovation**: The blockchain – a distributed ledger to record and secure transactions using cryptography.

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History of Blockchain

Blockchain1.0: Bitcoin (2008)



- **First Use Case**: Bitcoin was the first real-world application of blockchain, solving the "double-spending problem" (i.e., preventing someone from spending the same digital coin twice).

- **Proof of Work** (PoW): Introduced as the consensus mechanism to validate transactions and secure the network.

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History of Blockchain

Blockchain 2.0: Ethereum (2015)

- In 2015, **Vitalik Buterin** created **Ethereum**, a second-generation blockchain platform that expanded the use of blockchain beyond cryptocurrency.



- **Smart Contracts**: Ethereum introduced **programmable contracts** (smart contracts) that allow developers to build decentralized applications (dApps) on the blockchain.

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History of Blockchain

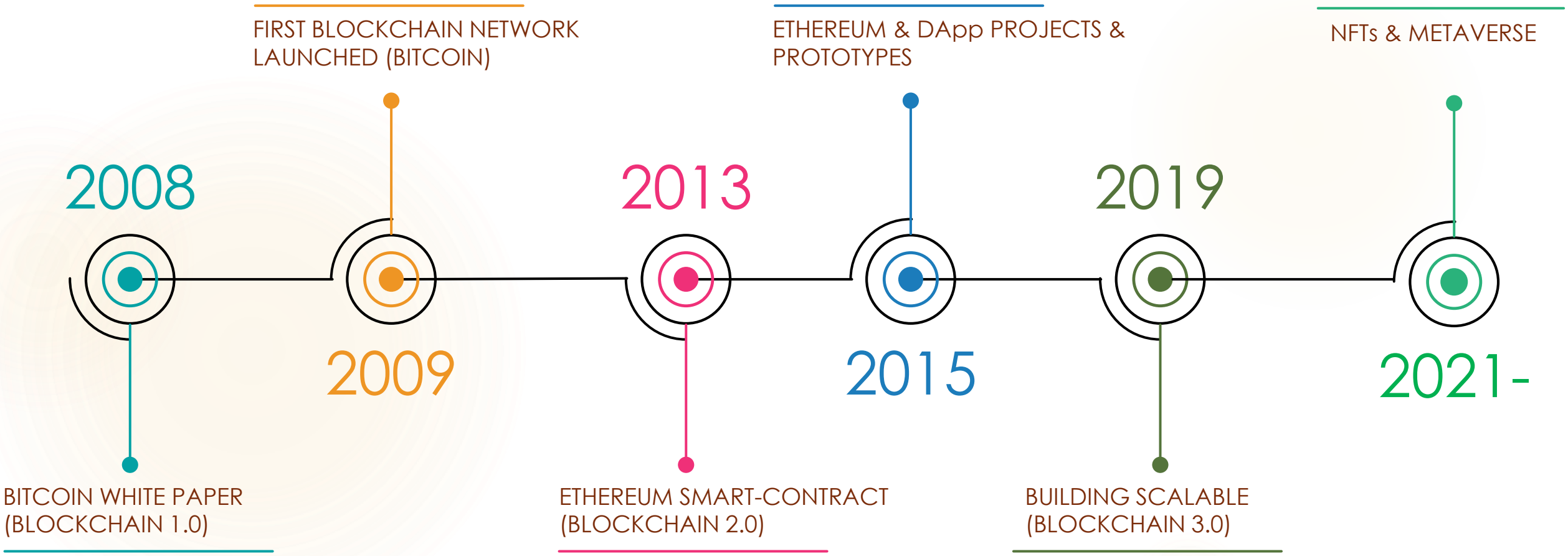
Evolution of Blockchain Use Cases

From Financial Transactions → Broader Applications

- While Bitcoin was focused solely on digital currency, Ethereum opened the door for blockchain to be used in **supply chain management**, **digital identity verification**, **voting systems**, **NFTs**, and more.
- Blockchain technology continues to evolve with new platforms and improvements in scalability, security, and usability (e.g., **Hyperledger**, **Cardano**, **Polkadot**).

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History of Blockchain



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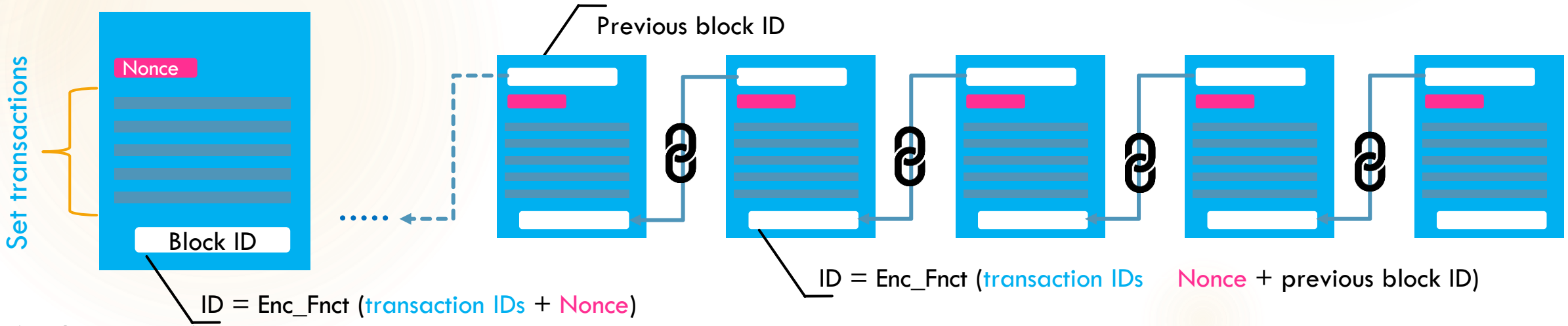
STRUCTURE OF A BLOCK IN BLOCKCHAIN

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Block-Chain Structure

What is a Block?

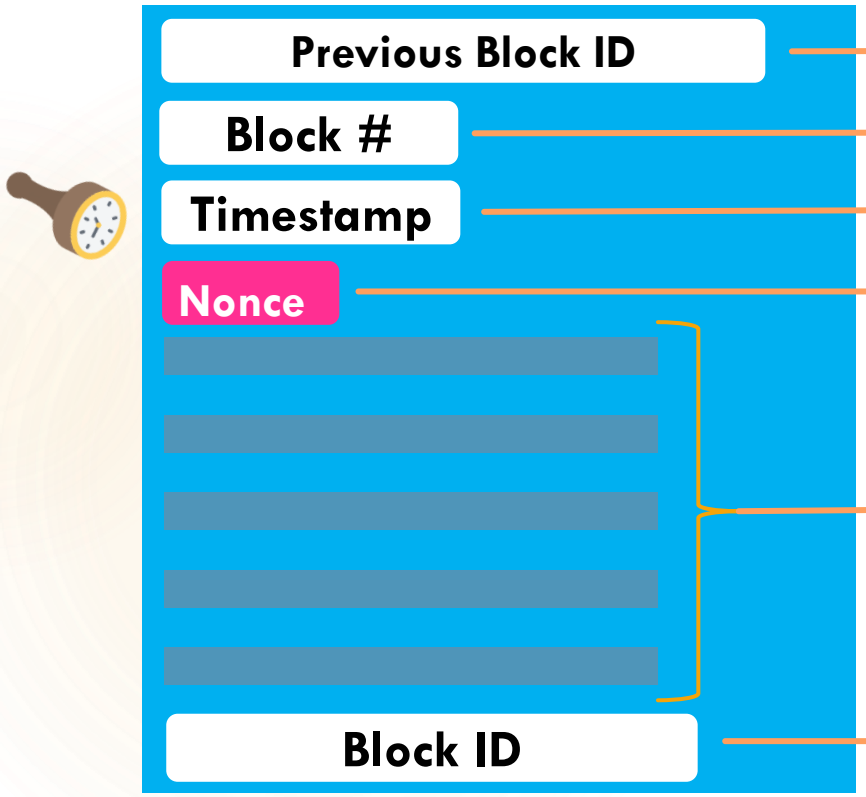
- A **block** is a fundamental unit of data in a blockchain.
- Each block stores information about transactions and links to the previous block, forming a chain.



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Block-Chain Structure

Components of a Block:



- The cryptographic hash of the previous block in the chain.
- The position of the block in the blockchain
- The approximate creation time of this block (Unix epoch time)
- The Proof of Work (PoW) or other consensus mechanism result.
- Set transactions
- The cryptographic hash generated based on the block's contents (transactions, timestamp, nonce, etc.).

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Block-Chain Structure

Components of a Block:

1. **Index**:
   - The position of the block in the blockchain (its number in the sequence).
2. **Timestamp**:
   - The exact time when the block was created.
   - Ensures that transactions are recorded chronologically.
3. **List of Transactions**:
   - The data stored in the block (e.g., transaction details, data records).
   - In Bitcoin, this would include information like the sender, receiver, and amount.
4. **Proof (Nonce)**:
   - The **Proof of Work** (PoW) or other consensus mechanism result.
   - This proves that computational work was done to create the block,

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Block-Chain Structure

Components of a Block:

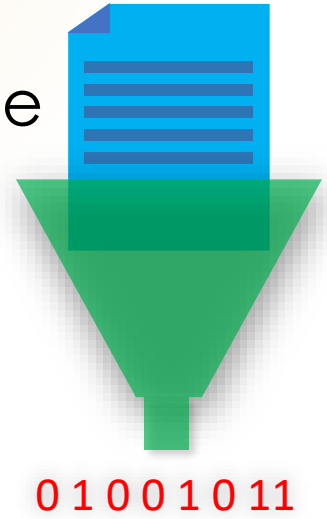
5. **Previous Hash**:
   - The cryptographic hash of the previous block in the chain.
   - This links the block to the one before it, ensuring that all blocks are connected.
   - **Benefit**: If anyone tries to alter a previous block, the hash will change, invalidating the entire chain.
6. **Current Block Hash**:
   - The cryptographic hash generated based on the block's contents (transactions, timestamp, nonce, etc.).
   - Ensures the integrity of the data within the block

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How Blocks are linked (Chained)

The Role of Cryptographic Hashing

- **Hash Function**:
  - A cryptographic hash function generates a fixed-size string (the hash) from input data (e.g., a block's contents).
  - The hash is like a **digital fingerprint**—any small change in the input data results in a completely different hash.



Enc_Fnct: Hash Function

Process of converting an **input** (any size) into a **fixed size** string.

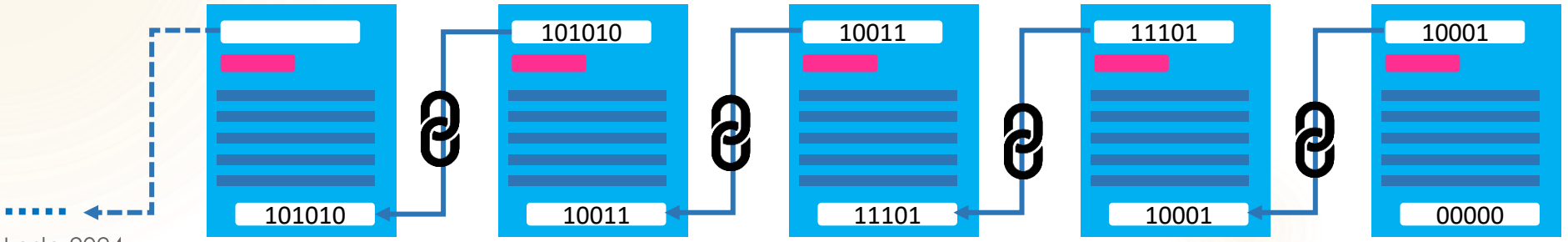
- **Unique:** different inputs produce different hash values..
- **Secure:** impossible to determine the input from its hash value.

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How Blocks are linked (Chained)

Linking Blocks Together

- **Previous Hash**:
  - Each block contains the **hash of the previous block**.
  - This connects the blocks together, forming a secure chain.
- **Current Hash**:
  - The current block's hash is generated based on its contents (transactions, timestamp, proof, and the previous block's hash).



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How Blocks are linked (Chained)

Why linking blocks is Important?

- **Tamper-Resistance**:
  - If a hacker tries to change the data in one block, the hash of that block will change.
  - Since the next block stores the previous block's hash, all subsequent blocks would also be affected, breaking the chain.
- **Security Through Consensus**:
  - Changing a block's hash would require re-mining all the following blocks, which is computationally expensive and impractical.
  - This ensures that the blockchain remains **immutable** and secure.

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TYPES OF BLOCKCHAIN NETWORKS

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Types of Blockchain Networks

(1) Public Blockchain

- **Open and Permissionless**:
  - A public blockchain is open to anyone who wants to participate in the network, validate transactions, or create new blocks.
  - **Examples**: Bitcoin, Ethereum.
- **Decentralization**:
  - Public blockchains are fully decentralized, with no central authority controlling the network.
- **Use Cases**:
  - Cryptocurrencies (Bitcoin), decentralized applications (Ethereum), and peer-to-peer transactions.

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Types of Blockchain Networks

(2) Private Blockchain

- **Restricted Access**:
  - Private blockchains are closed and accessible only to a specific group of participants, typically within an organization.
  - **Examples**: Hyperledger, Corda.
- **Centralized Governance**:
  - Controlled by a single organization or a group of entities, offering more control over who can read, write, or validate data on the blockchain.
- **Use Cases**:
  - Enterprise applications like supply chain management, internal financial systems, and identity management.

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Types of Blockchain Networks

(3) Consortium Blockchain

- **Partially Decentralized**:
  - A consortium blockchain is controlled by a group of organizations rather than a single entity.
  - **Examples**: R3 Corda, Quorum.
- **Permissioned Participation**:
  - Only approved participants can join the network, but multiple organizations share control.
- **Use Cases**:
  - Industry-wide applications, such as inter-bank transfers, consortiums in trade finance, and collaborative business networks.

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Types of Blockchain Networks

Type	Access	Governance	Use Cases
Public	Open to all	Decentralized	Cryptocurrencies, dApps
Private	Restricted (permissioned)	Centralized	Enterprise, internal systems
Consortium	Restricted (permissioned)	Partially decentralized	Industry collaborations, finance

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Blockchain Applications

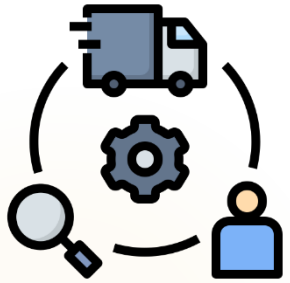
Cryptocurrencies (Bitcoin, Ethereum)



- **Digital Currency**:
  - Bitcoin and Ethereum are the most popular examples of blockchain being used for decentralized digital currencies.
  - **Eliminating Intermediaries**: Transactions can be made without the need for banks or payment processors.
- **Key Benefit**: Fast, secure, cheap, and decentralized transactions without third-party involvement.

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Blockchain Applications

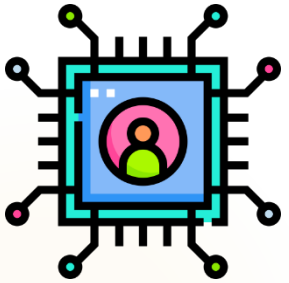


Supply Chain Management

- **Improving Transparency**:
  - Blockchain provides real-time tracking of goods and products throughout the supply chain.
  - **Example**: Companies like Walmart and IBM are using blockchain to trace the origin of food products from farm to table.
- **Key Benefit**: Enhanced traceability, fraud reduction, and efficient management of goods.

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Blockchain Applications



Digital Identity Verification

- **Secure Identity Management**:
  - Blockchain can be used to create secure, verifiable digital identities that are controlled by the individual.
  - **Example**: Projects like **ID2020** use blockchain to provide individuals with secure digital identities that can be used for banking, healthcare, and more.
- **Key Benefit**: Enhanced privacy and control over personal data.

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Blockchain Applications

Voting Systems



- **Tamper-Proof Voting:**
  - Blockchain can be used to create transparent, secure, and tamper-proof voting systems.
  - Example: Countries like Estonia have experimented with blockchain-based voting systems to ensure election integrity.
- **Key Benefit:** Increased transparency and reduced risk of voter fraud.

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Blockchain Applications

Healthcare



- **Patient Data Management**:
  - Blockchain can be used to securely store and share patient data, for better coordination between healthcare providers.
  - **Example**: Projects like **Medicalchain** use blockchain to store medical records securely, ensuring that only authorized individuals can access the data.
- **Key Benefit**: Improved data security, patient control over medical information, and better collaboration among healthcare providers.

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Blockchain Applications

Real-World Examples Table:

Use Case	Example	Key Benefit
Cryptocurrencies	Bitcoin, Ether	Secure, decentralized transactions
Supply Chain Management	IBM Food Trust, Walmart	Transparency, fraud reduction
Digital Identity	ID2020	Privacy, data control
Voting	Estonia blockchain voting experiment	Tamper-proof voting
Healthcare	Medicalchain	Secure, shareable medical records

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BLOCKCHAIN is a revolutionary technology that addresses **trust issues** in transactional systems, providing a **secure, decentralized,** and **transparent** way to manage data and transactions across multiple industries.

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